B.4.3 Logging

Whether you realize it or not, you've been using Django's logging functionality all along. The development server console messages indicating a view was visited are from the logging system. Django's logging is built on top of Python's logging, which is extremely configurable (read: a bit complicated to get going).

Python breaks the logging system down into four parts:

- Loggers
- Filters
- Handlers
- Formatters

The logger is what you interact with to log a message. The logger object has a variety of methods that correspond to the log level of the message you are logging. The log levels are

- DEBUG
- INFO
- WARNING
- ERROR
- CRITICAL

The logging filter is responsible for determining which messages get recorded in the logs. The simplest filters are based on the log level. If your filter is set to ERROR, only messages marked as ERROR or CRITICAL get recorded. You likely want different filters in different environments. In development, you want everything, which you can get by setting the log level to DEBUG. In production, that's a bit too noisy, and you may set it higher. The WARNING level is pretty typical on a production server. If a problem occurs in production, you might temporarily change the log level to capture information and help you debug the issue.

The logging handler determines where a message gets logged. The two most common destinations are the console and a file. A logging formatter specifies the format of the message when it is logged. In addition to the message getting logged, you can also add information, such as the date and time, the called function, the process and thread IDs, the source code line, and the current phase of the moon. (That last one is a joke, but there are over 20 log attributes you can choose from: https://docs.python.org/3/library/logging.html#logrecord-attributes). You specify the format using a `%s` style string that names the attributes. For example, `"%(message)s"` is the format for logging only the message.

Django's own code uses different logging objects for different kinds of messages. The `django.request` logger logs the messages you've seen when using the development server. The `django.db.backends` logger is for database interactions. This classification allows you to determine which messages make it into your logs. It is also hierarchical; configuring for the `django` logger gets both the `.request` and `.db.backends` content, along with everything else.